Pain Ther
https://doi.org/10.1007/s40122-026-00902-2

ORIGINAL RESEARCH

# Pudendal Neuralgia in a Tertiary Pain Clinic: Presentation, Treatment, and Factors Associated with Neuromodulation in 224 Adults

Paz Kelmer · Shachar Zion Shemesh · Philip Rossovsky · Yotam Hadari · Itay Goor-Aryeh · Lior Ungar

Received: September 3, 2026 / Accepted: October 21, 2026
© The Author(s) 2026

## ABSTRACT

***Introduction*:** Pudendal neuralgia is an underrecognized cause of chronic, disabling neuropathic perineal pain affecting both women and men. Real-world data on its presentation, its treatment within a specialty pain service, and the factors that accompany escalation to interventional therapy are scarce. We hypothesized that an iatrogenic (postsurgical) origin and relief after a diagnostic pudendal nerve block would identify the patients who reach neuromodulation.

***Methods*:** We performed a retrospective manual review of the complete electronic records of all adults with documented pudendal neuralgia, pudendal neuropathy, or pudendal nerve entrapment managed at a single tertiary pain clinic between 2009 and 2026. Demographics, clinical presentation, presumed etiology, treatments, and outcomes were abstracted from the free-text record. Ascertainment was in two stages: variables were first identified by structured text screening of the free-text record, and the two variables most susceptible to false-positive screening, a postsurgical etiology and receipt of neuromodulation, were then verified against the source charts by manual review of every screen-positive record. The chart-verified definitions are used throughout. Associations between binary variables were tested by the chi-squared test, with odds ratios (OR) and 95% confidence intervals (CI); factors associated with neuromodulation were assessed by multivariable logistic regression.

***Results*:** Among 224 patients (138 women [61.6%], 86 men [38.4%]; median age 48.5 years, interquartile range 35–64), pain was worse when sitting in 57.6% and burning in 39.3%, and involvement was bilateral in 21.9%. Text screening identified a possible postsurgical onset in 46.4%, but chart re-adjudication confirmed a named operation preceding the pain in only 31 patients (13.8%), of whom 6 had undergone a clearly non-pelvic procedure; confirmed postsurgical onset was therefore less frequent than documented entrapment features (24.1%) and comparable to childbirth (15.9% of women). Care was heavily interventional: 59.4% received a pudendal nerve block, 28.6% pulsed

P. Kelmer (✉) · S. Z. Shemesh · P. Rossovsky · L. Ungar
Department of Neurosurgery, Sheba Medical Center, Ramat Gan, Israel
e-mail: pazkelmer92@gmail.com

P. Kelmer · S. Z. Shemesh · Y. Hadari
Gray Faculty of Medicine, Tel Aviv University, Tel Aviv, Israel

I. Goor-Aryeh · L. Ungar
Pain Center, Sheba Medical Center, Ramat Gan, Israel

L. Ungar
Dina Recanati School of Medicine, Reichman University, Herzliya, Israel

Published online: 07 November 2026

radiofrequency, and 9.4% underwent surgical decompression. Neuromodulation was documented in the record of 64 patients (28.6%), but an electrode was placed in only 26 (11.6%) and a permanent device implanted in 18 (8.0%): spinal cord stimulation in 14, sacral neuromodulation in 2, dorsal root ganglion stimulation in 1, and pudendal nerve stimulation in 1. Three factors were independently associated with documented neuromodulation: relief after a diagnostic block (adjusted OR 4.88, 95% CI 1.91–12.46%), a refractory course (2.92, 1.52–5.60%), and chart-confirmed postsurgical onset (2.83, 1.17–6.82%); *c*-statistic 0.76. In an exploratory model restricted to the 18 implanted patients, a refractory course (7.40, 1.98–27.61%) and postsurgical onset (4.24, 1.29–13.94%) remained associated with the outcome whereas relief after a block did not (5.90, 0.72–48.31%).

***Conclusions***: In routine pain-clinic practice, pudendal neuralgia is managed along an interventional pathway, but device implantation is considerably less common than the frequency with which neuromodulation is discussed. A confirmed postsurgical onset is less common than text screening suggests, yet is disproportionately associated with escalation. Being observational, these findings describe how care proceeds rather than establishing cause.

## PLAIN LANGUAGE SUMMARY

Pudendal neuralgia is a long-lasting pain in the area between the legs, caused by irritation or injury of the pudendal nerve. It affects women and men, is often worse when sitting, and is not well known, so people may wait a long time before anyone identifies the cause. We looked back at the records of 224 adults treated for this condition at a specialist pain clinic in Israel over 16 years, to learn what brought the pain on and which treatments people received. When we first searched the records by computer, it looked as though almost half had developed pain after an operation, but when we read each chart by hand, we could confirm an operation before the pain in about one in seven. Reading the charts also changed what we found about implanted devices: the records mentioned this treatment for about 1 in 4 people, but a device was actually fitted in about 1 in 12, and most of those were spinal cord stimulators. Most people were treated with injections that numb the nerve. Those whose pain had not responded to earlier treatments, and those whose pain began after an operation, were the most likely to go on to a device. Pain relief after a numbing injection went together with a device being discussed, but not clearly with one being fitted. Because we looked back at existing records rather than running a trial, our findings describe what happened in practice rather than proving what causes what.

**Keywords:** Chronic pelvic pain; Nerve block; Neuromodulation; Neuropathic pain; Pudendal nerve; Pudendal neuralgia

### Key Summary Points

*Why carry out this study?*

Pudendal neuralgia is an under-recognized cause of chronic, disabling neuropathic perineal pain in both women and men, and real-world descriptions of how it presents and is managed within specialty pain services are scarce.

We asked which patients escalate to neuromodulation, hypothesizing that an iatrogenic (postsurgical) origin and relief after a diagnostic pudendal nerve block would identify that group.

*What was learned from the study?*

Among 224 adults, chart verification confirmed a named operation preceding the pain in 31 (13.8%), fewer than documented entrapment features (24.1%); text screening alone would have indicated 46.4%.

Neuromodulation was documented in the records of 64 patients (28.6%), but an electrode was placed in 26 (11.6%) and a permanent device implanted in 18 (8.0%); spinal cord stimulation accounted for 14 of the 18 devices, with sacral neuromodulation in 2, dorsal root ganglion stimulation in 1, and pudendal nerve stimulation in 1.

Relief after a diagnostic block (adjusted odds ratio 4.88), a refractory course (2.92), and chart-confirmed postsurgical onset (2.83) were independently associated with documented neuromodulation; in an exploratory analysis of the 18 implanted patients, refractoriness and postsurgical onset persisted but relief after a block did not.

Because block response and refractoriness themselves inform the decision to offer neuromodulation, these associations describe the care pathway rather than predict which patients will escalate; and in routine records, a documented treatment is not evidence that the treatment was delivered.

## INTRODUCTION

Pudendal neuralgia is a neuropathic pain syndrome arising in the sensory territory of the pudendal nerve, the clitoris or penis, vulva or scrotum, perineum, and anorectum, and is an important though frequently overlooked contributor to chronic pelvic and perineal pain in both women and men [1–5]. The pudendal nerve is vulnerable to compression along its course between the sacrotuberous and sacrospinous ligaments and within Alcock's (pudendal) canal, the anatomical basis for entrapment described in cadaveric and surgical work [6–8]. Because chronic pain is now recognized as a disease in its own right and neuropathic pain has an explicit diagnostic and therapeutic framework [9, 10], accurate characterization of pudendal neuralgia has direct management implications.

The diagnosis remains essentially clinical. The Nantes criteria specify pain in the pudendal territory, worsened by sitting, without objective sensory loss and without nocturnal awakening, together with a positive response to an anesthetic pudendal nerve block [3]. Etiologies range from mechanical entrapment to obstetric injury and iatrogenic damage following pelvic surgery, including mesh and sling procedures, yet the relative frequency of these mechanisms in a specialty pain population, and the way they shape treatment, are not well described [4, 5, 11, 12]. Management spans a broad ladder: education and pelvic-floor physiotherapy, neuropathic-pain pharmacotherapy, image-guided pudendal nerve blocks, pulsed radiofrequency, botulinum toxin, neuromodulation, and surgical decompression [13–26]. Which patients traverse that ladder to its most invasive options is uncertain, and this matters because neuromodulation is resource-intensive and irreversible in practice, so understanding who reaches it informs both counseling and service planning.

We therefore performed a retrospective manual medical-record review of all adults with pudendal neuralgia managed at a tertiary pain clinic, with the aims of characterizing their presentation, presumed etiology, and the full spectrum of treatments received, and of identifying the factors associated with escalation to neuromodulation. We hypothesized that an iatrogenic (postsurgical) etiology and a positive diagnostic pudendal nerve block, rather than demographic features such as sex, would be independently associated with receiving neuromodulation.

## METHODS

### Study Design and Population

This was a retrospective study based on a manual review of complete electronic medical records at a single tertiary specialty pain clinic. Records were identified by searching the clinic database for documented pudendal neuralgia, pudendal neuropathy, or pudendal nerve entrapment. Of 549 candidate records retrieved for the period 2009–2026, 324 (59.0%) were excluded on manual review and 1 further patient younger than 18 years was excluded, leaving 224 adults for analysis. The 324 exclusions fell into three groups:

records in which the search term appeared only as a differential diagnosis that was subsequently discounted, or as a quotation from an outside referral letter, without the treating clinician ever recording the diagnosis; records in which the term appeared in a template, a past-history list, or a discussion of another patient's condition; and records belonging to patients assessed for pelvic or perineal pain in which an alternative diagnosis (for example coccydynia, endometriosis, interstitial cystitis, or lumbosacral radiculopathy) was recorded instead. We did not require the diagnosis to satisfy the Nantes criteria, only that a treating clinician had recorded it; the proportion in each exclusion group was not tabulated at the time of review and cannot be reconstructed retrospectively, which we acknowledge as a limitation.

For each patient, the full record was read to abstract demographics, the clinical features and laterality of the pain, the presumed etiology, the treatments received (conservative, pharmacological, interventional, neuromodulatory, and surgical), documented responses, and the clinical course. The primary review was performed by three of the authors (P.K., S.Z.S., P.R.), each of whom read the complete record for every assigned patient; cases in which the abstraction was uncertain were resolved by discussion, and any case still unresolved was referred to the senior pain physician supervising the study (L.U.), whose judgment was final. The same arrangement was used for the chart verification of the postsurgical and neuromodulation variables described below. A feature was coded as present only when explicitly documented. Data were deidentified and ages were recorded to the nearest year.

## Definitions

Pain worse when sitting, burning quality, allodynia or foreign-body sensation, and laterality were recorded when explicitly documented. A postsurgical (iatrogenic) etiology was initially identified by text screening for documentation of pain onset after a pelvic or perineal operation. Because such screening cannot distinguish an operation that preceded and plausibly caused the pain from an operation mentioned elsewhere in the history, every screen-positive record was re-opened and verified against the chart, and the treating clinician's named operation, its anatomical site, and its temporal relation to pain onset were recorded. A postsurgical etiology was confirmed only where a specific operation was named and recorded as preceding the pain; where the operation named was clearly nonpelvic (for example lumbar discectomy or limb fracture fixation), the patient was not counted as postsurgical in the primary analysis. We note explicitly that temporal sequence does not establish iatrogenic nerve injury, and the confirmed variable should be read as postsurgical onset rather than as proven operative trauma. Entrapment was recorded when entrapment, Alcock-canal involvement, or the corresponding surgical indication was documented, and post-childbirth onset when the pain was attributed to vaginal delivery. Documented symptomatic relief after a block denoted any physician-recorded improvement in pain following a nerve block or injection. No numerical threshold, such as at least 50% relief, could be applied, because a quantified pre- and post-block pain score was recorded for only a small minority of blocks; the variable is therefore a qualitative clinical judgment made by the treating physician and recorded in the note, not a standardized responder definition. Two consequences follow. First, the variable is susceptible to interpretation bias, since what one clinician records as improvement another may not record at all. Second, absence of a recorded response was coded as absence of documented relief and cannot be distinguished from a response that occurred but was not written down; the proportion with documented relief is therefore a lower bound. Because a block response is also one of the clinical reasons neuromodulation is offered, we treat its association with neuromodulation as a description of the care pathway rather than as a predictive finding, and we examined it separately against documented neuromodulation and against actual device implantation. A refractory course denoted explicit documentation of treatment resistance.

Neuromodulation required particular care. Text screening identifies any mention of spinal

cord, dorsal root ganglion, sacral, or pudendal nerve stimulation, including mentions that record a discussion, a referral, a declined offer, or a plan that was never executed. Every screen-positive record was therefore re-opened and the charts read specifically to establish which device, if any, was placed, the date, and whether only a trial electrode was sited without a permanent implant. Three nested outcomes were defined: neuromodulation documented anywhere in the record; an electrode placed, whether trial or permanent; and a permanent device implanted. The first is the primary outcome, because it corresponds to the clinical decision point at which neuromodulation enters the care pathway and because it provides the event count needed for a five-variable model. The third is reported as a prespecified secondary outcome and, with 18 events, is explicitly exploratory.

### Sex and Gender Considerations

Pudendal neuralgia affects both women and men, and a female predominance and an obstetric etiology unique to women are both described in the literature [3–5, 11]. In accordance with the Sex and Gender Equity in Research (SAGER) guidelines, sex was recorded for every patient as the biological attribute documented in the medical record; gender identity was not recorded in routine notes and could not be analyzed. Results are disaggregated by sex where the distinction is informative, post-childbirth onset is reported as a proportion of women rather than of the whole cohort, and female sex was entered as a covariate in the multivariable model.

### Statistical Analysis

Continuous variables are summarized as median and interquartile range (IQR) and categorical variables as counts and percentages. Associations between binary variables were tested with the chi-squared test, with odds ratios (OR) and 95% confidence intervals (CI). Factors associated with escalation to neuromodulation were examined with a multivariable logistic-regression model including chart-confirmed postsurgical etiology, documented relief after a diagnostic block, a refractory course, bilateral involvement, and female sex; with 64 events for the primary outcome this corresponded to approximately 13 events per variable. The identical model was refitted with permanent device implantation as the outcome; with 18 events, this gives approximately 3.6 events per variable, well below conventional requirements, and that model is reported as exploratory with correspondingly wide confidence intervals. Model discrimination was assessed with the *c*-statistic (area under the receiver-operating-characteristic curve) and calibration with the Hosmer–Lemeshow test.

Because features were coded from the free-text record, a feature was recorded as present only when explicitly documented, and absence of documentation was coded as absence of the feature. This convention means that every reported proportion is a lower bound on true prevalence and that true absence cannot be distinguished from absent documentation; it applies to the clinical presentation features, the etiologic categories, and the recorded treatment responses. The outcome variables and the postsurgical variable are not subject to this limitation in the same way, because both were verified against the source charts. Age was recovered from the charts where it had not been captured by automated extraction and was available for all 224 patients, as was sex. Age was not used as a model covariate. There were therefore no missing values among analyzed variables and no imputation was performed. Robustness was assessed in two prespecified ways: adjustment for treatment era (index year on or after 2021) and an analysis restricted to patients with above-median documentation density.

Two-sided *P* values are reported and statistical significance was set at $P < 0.05$. Because multiple associations were examined and ascertainment was based on documented rather than protocolized findings, we emphasize effect sizes over *P* values and did not formally adjust for multiplicity. Analyses were performed in Python 3 with pandas and NumPy; logistic regression was implemented by iteratively reweighted least squares and verified against the closed-form odds ratio and standard error of saturated two-by-two tables.

### Compliance with Ethical Guidelines

The study was approved by the institutional Helsinki (research ethics) committee of the Sheba Medical Center, Tel HaShomer, Israel (approval no. SMC-8148-21; Israeli Ministry of Health no. 202121003), with a waiver of informed consent for this retrospective analysis of de-identified records. It was performed in accordance with the Declaration of Helsinki of 1964 and its later amendments. No individually identifiable data are presented, and consent for publication was therefore not applicable. Patients were not involved in the design, conduct, or reporting of this retrospective record-based study.

## RESULTS

### Cohort and Presentation

We identified 224 patients, of whom 138 (61.6%) were women and 86 (38.4%) men; the median age was 48.5 years (IQR 35–64). Pain was documented as worse when sitting in 129 patients (57.6%) and had a burning or neuropathic quality in 88 (39.3%); allodynia or a foreign-body sensation was recorded in 23 (10.3%) and involvement was bilateral in 49 (21.9%) (Table 1; Fig. 1).

These proportions are lower than those reported in series that apply the Nantes criteria prospectively, and the reason is methodological rather than clinical. Explicit reference to the Nantes criteria was uncommon in routine notes, and features were coded only when a clinician had written them down. Sitting-related pain in particular is so characteristic of the diagnosis that it may be assumed rather than restated at every visit. The figures should therefore be read as the proportion in whom the feature was documented, not the proportion in whom it was present, and this cohort is clinician-defined rather than Nantes-defined.

**Table 1** Cohort characteristics and clinical presentation ($n = 224$)

| Characteristic | Value |
|---|---|
| Female, *n* (%) | 138 (61.6) |
| Male, *n* (%) | 86 (38.4) |
| Age, years, median [IQR] | 48 [35–63] |
| Pain worse when sitting, *n* (%) | 129 (57.6) |
| Burning/neuropathic quality, *n* (%) | 88 (39.3) |
| Bilateral involvement, *n* (%) | 49 (21.9) |
| Allodynia/foreign-body sensation, *n* (%) | 23 (10.3) |
| Postsurgical (iatrogenic) onset, *n* (%) | 104 (46.4) |
| Entrapment/Alcock canal, *n* (%) | 54 (24.1) |
| Postsurgical onset, text screening, *n* (%) | 104 (46.4) |
| Postsurgical onset, chart-confirmed, *n* (%) | 31 (13.8) |
| Post-childbirth onset, *n* (% of women) | 22 (15.9) |

Features reflect documentation in the free-text record and are coded as present only when explicitly recorded, so each proportion is a lower bound on true prevalence. Post-childbirth onset is expressed as a proportion of the 138 women. The two postsurgical rows are the same patients assessed by two methods: text screening of the free-text record, and chart verification of every screen-positive record, requiring a named operation preceding the pain at a pelvic, perineal, or abdominal site. The chart-confirmed definition is used in all analyses

*IQR* interquartile range

### Etiology

Text screening flagged a possible postsurgical onset in 104 patients (46.4%). On chart re-adjudication of all 104, a specific operation preceding the pain was named in 37; in the remaining 67, no qualifying operation could be identified, the screening term having arisen from unrelated surgical history, a planned future operation, or a template entry. Of the

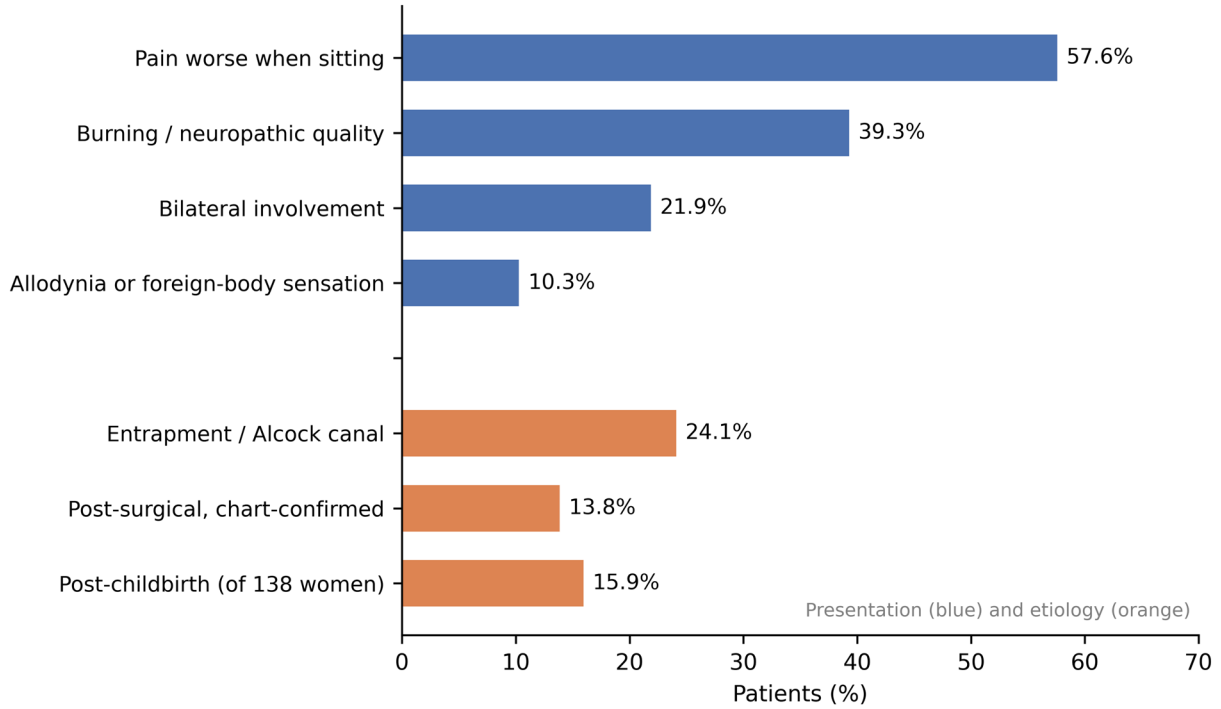



**Fig. 1** Clinical presentation (blue) and etiology (orange) of pudendal neuralgia ($n$ = 224). Bars show the proportion of patients with each documented feature. Pain worse when sitting was the most frequently documented presentation. Among etiologies, documented entrapment features were more frequent than a chart-confirmed post-surgical onset. Features were coded only when explicitly recorded, so proportions are lower bounds. Post-childbirth onset is expressed as a proportion of the 138 women

37 named operations, 6 were clearly nonpelvic (lumbar discectomy in two, spinal decompression, hip labral surgery, femoral fracture fixation, and hand extensor tendon repair), leaving 31 patients (13.8%) with a confirmed pelvic, perineal, or abdominal operation preceding the pain. The named procedures were other pelvic or perineal operations in 8, hysterectomy in 5, colorectal surgery in 5, endometriosis laparoscopy in 5, prolapse repair in 3, hernia repair in 3, prostatectomy in 1, and an obstetric operation in 1; the recorded site was pelvic in 20, perineal in 7, and abdominal in 4.

Confirmed postsurgical onset was therefore less frequent than documented entrapment features, which were recorded in 54 patients (24.1%), and of similar frequency to childbirth, to which pain was attributed in 22 women (15.9% of women). Screening alone would have placed postsurgical onset first among the identifiable mechanisms; chart verification places it third.

## Treatments

Care spanned the full therapeutic ladder (Table 2; Fig. 2). Conservative pelvic-floor physiotherapy was used in 52.7%. Neuropathic-pain pharmacotherapy was common: pregabalin in 61.2%, duloxetine in 41.1%, amitriptyline in 29.0%, and gabapentin in 21.0%, while 32.1% received opioids and 37.5% medical cannabis. Interventional therapy was central: 59.4% underwent a pudendal nerve block (imaging-guided in 56.2% of the cohort), 32.1% a ganglion impar block, 28.6% pulsed radiofrequency, 21.0% radiofrequency ablation, and 10.7% botulinum toxin. Surgical decompression was performed in 9.4%.

**Table 2** Treatment utilization and outcomes ($n = 224$)

| Modality/outcome | *n* (%) |
|---|---|
| Pelvic-floor physiotherapy | 118 (52.7) |
| Pregabalin | 137 (61.2) |
| Duloxetine | 92 (41.1) |
| Amitriptyline | 65 (29.0) |
| Gabapentin | 47 (21.0) |
| Opioids | 72 (32.1) |
| Medical cannabis | 84 (37.5) |
| Pudendal nerve block | 133 (59.4) |
| of which imaging-guided[a] | 126 (56.2) |
| Ganglion impar block | 72 (32.1) |
| Pulsed radiofrequency | 64 (28.6) |
| Radiofrequency ablation | 47 (21.0) |
| Botulinum toxin | 24 (10.7) |
| Neuromodulation documented in the record | 64 (28.6) |
| Electrode placed (trial or permanent) | 26 (11.6) |
| Permanent device implanted | 18 (8.0) |
| Spinal cord stimulation (SCS) | 14 (6.3) |
| Sacral neuromodulation (SNM) | 2 (0.9) |
| Dorsal root ganglion (DRG) stimulation | 1 (0.4) |
| Peripheral nerve stimulation, sciatic | 1 (0.4) |
| Dedicated pudendal nerve stimulation | 0 (0) |
| Surgical decompression | 21 (9.4) |
| Documented symptomatic relief after block | 154 (68.8) |
| Refractory course | 91 (40.6) |

*DRG* dorsal root ganglion, *SCS* spinal cord stimulation, *SNM* sacral neuromodulation

[a]Percentage is of the whole cohort ($n = 224$), not of those who received a block

[b]The three neuromodulation rows are nested: every patient with an electrode placed is also counted in the documented row, and every implanted patient in both rows above. Of the 64 with neuromodulation documented, 8 underwent a trial electrode without a permanent implant and 38 had no electrode placed at any time. Four of the 18 permanent devices were explanted during follow-up (inadequate efficacy in 2, conversion to deep brain stimulation in 1, reason not recorded in 1). Modality was established from the operative and procedural records of every patient in whom neuromodulation was documented

Neuromodulation requires separate reporting (Fig. 2b). It was documented somewhere in the record of 64 patients (28.6%), but chart review established that an electrode was placed in only 26 (11.6%) and that a permanent device was implanted in 18 (8.0%). Of the remaining 46, 8 underwent a trial electrode that was removed without a permanent implant, and in 38 no electrode was ever placed: the record documented a discussion, a referral, a declined offer, or a plan that was not executed. Among the 18 implanted devices, spinal cord stimulation accounted for 14, sacral neuromodulation for 2, and dorsal root ganglion stimulation for 1. The remaining device was recorded in the procedure history as spinal cord stimulation but the operative description established a peripheral nerve stimulator placed along the sciatic nerve; no patient in this cohort received a dedicated pudendal nerve stimulator. Approximately 4 of the 18 permanent devices were explanted during follow-up, for inadequate efficacy in 2, conversion to deep brain stimulation in 1, and for a reason not recorded in 1.

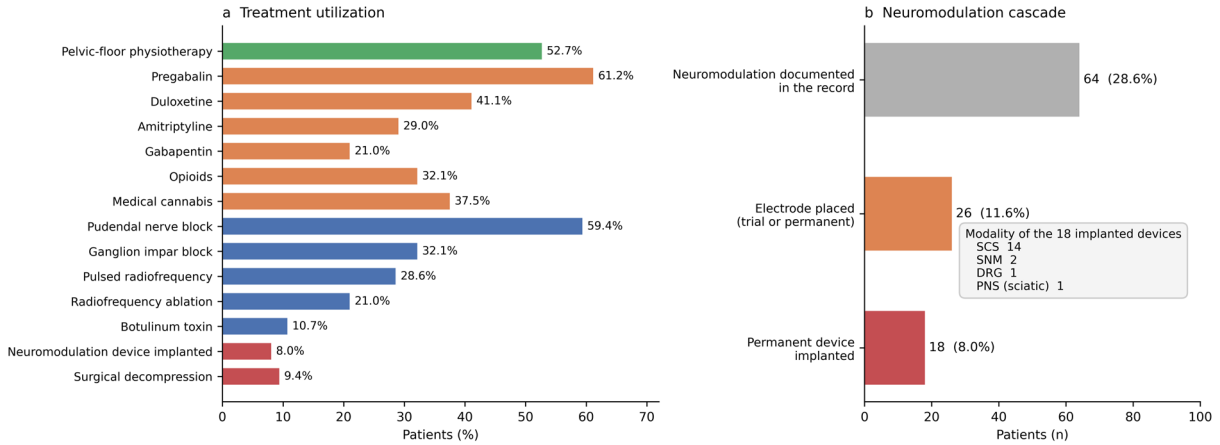



**Fig. 2** Treatment utilization ($n$ = 224). **a** Proportion of patients receiving each modality, grouped as conservative (green), pharmacological (orange), interventional (blue), and device or surgery (red). **b** The neuromodulation cascade: neuromodulation was documented in the records of 64 patients, an electrode was placed in 26, and a permanent device was implanted in 18. The inset gives the modality of the 18 implanted devices. *SCS* spinal cord stimulation, *SNM* sacral neuromodulation, *DRG* dorsal root ganglion stimulation, *PNS* peripheral nerve stimulation

## Outcomes and Factors Associated with Escalation to Neuromodulation

Symptomatic relief after a block or injection was documented in 68.8% of patients, while 40.6% were described as refractory. Chart-confirmed postsurgical onset was associated with escalation: compared with patients without a confirmed operation, these patients more often had neuromodulation documented (OR 2.36, 95% CI 1.08–5.12%) and more often received a permanent device (OR 3.62, 95% CI 1.25–10.51%).

In the multivariable model for documented neuromodulation (Table 3; Fig. 3), three factors were independently associated with the outcome: documented relief after a diagnostic block (adjusted OR 4.88, 95% CI 1.91–12.46%; $P$ = 0.001), a refractory course (2.92, 1.52–5.60%; $P$ = 0.001), and chart-confirmed postsurgical onset (2.83, 1.17–6.82%; $P$ = 0.021). Bilateral involvement was of borderline significance (2.10, 1.01–4.36%; $P$ = 0.047) and female sex was not associated with the outcome (1.09, 0.55–2.16%; $P$ = 0.80). The model showed acceptable discrimination ($c$-statistic 0.76) and adequate calibration (Hosmer–Lemeshow $P$ = 0.84).

In the exploratory model restricted to the 18 patients who received a permanent device, the ordering changed. A refractory course (adjusted OR 7.40, 95% CI 1.98–27.61%; $P$ = 0.003) and chart-confirmed postsurgical onset (4.24, 1.29–13.94%; $P$ = 0.017) remained associated with the outcome and were numerically stronger, whereas documented relief after a block did not reach significance (5.90, 0.72–48.31%; $P$ = 0.10). Confidence intervals are wide and the analysis is underpowered at 3.6 events per variable, so this contrast should be regarded as hypothesis-generating. It nonetheless suggests that a documented block response accompanies the decision to consider neuromodulation more closely than it accompanies the decision to implant.

Neuromodulation was documented in 41.3% of patients first seen in 2010–2020 versus 19.7% of those first seen in 2021–2026. We interpret this difference as largely an artifact of unequal follow-up opportunity rather than as a change in practice: escalation to neuromodulation typically follows a sequence of conservative, pharmacological, and interventional steps that takes years, so patients entering the service later have had less time to traverse it. A changing treatment strategy cannot be excluded and may contribute, since the availability of dorsal root ganglion and dedicated pudendal stimulation increased over the study period while referral thresholds may

**Table 3** Factors associated with escalation to neuromodulation, under two outcome definitions

| Factor | Primary outcome: neuromodulation documented (64 events) | | Secondary outcome: permanent device implanted (18 events, exploratory) | |
|---|---|---|---|---|
| | Unadjusted OR (95% CI) | Adjusted OR (95% CI); *P* | Unadjusted OR (95% CI) | Adjusted OR (95% CI); *P* |
| Documented relief after diagnostic block | 6.44 (2.63–15.82%) | 4.88 (1.91–12.46%); 0.001 | 8.56 (1.12–65.68%) | 5.90 (0.72–48.31%); 0.098 |
| Refractory course | 3.56 (1.94–6.53%) | 2.92 (1.52–5.60%); 0.001 | 8.55 (2.40–30.50%) | 7.40 (1.98–27.61%); 0.003 |
| Postsurgical onset, chart-confirmed | 2.36 (1.08–5.12%) | 2.83 (1.17–6.82%); 0.021 | 3.62 (1.25–10.51%) | 4.24 (1.29–13.94%); 0.017 |
| Bilateral involvement | 3.24 (1.67–6.28%) | 2.10 (1.01–4.36%); 0.047 | 2.48 (0.91–6.80%) | 1.54 (0.51–4.67%); 0.448 |
| Female sex | 1.40 (0.76–2.58%) | 1.09 (0.55–2.16%); 0.799 | 0.98 (0.36–2.63%) | 0.70 (0.24–2.05%); 0.512 |

Each set of adjusted estimates comes from a single multivariable logistic-regression model containing all five factors. Primary model: 64 events, approximately 13 events per variable, *c*-statistic 0.76, Hosmer–Lemeshow $P = 0.84$. Secondary model: 18 events, approximately 3.6 events per variable, *c*-statistic 0.83; this model is underpowered and is reported as exploratory

*CI* confidence interval, *OR* odds ratio

also have shifted, but these data cannot separate the two explanations: we have no fixed-horizon outcome because the date of each decision is not uniformly recorded. The associations were therefore additionally adjusted for treatment era, after which all three persisted, although the estimate for postsurgical onset became borderline (postsurgical 2.51, 95% CI 1.02–6.20%; block relief 4.48, 1.74–11.48%; refractory 2.78, 1.44–5.36%). Permanent implantation showed the same direction across eras (10.9% versus 6.1%).

A second prespecified sensitivity analysis restricted the cohort to the 109 patients with above-median documentation density. Here, only a refractory course remained independently associated with documented neuromodulation (adjusted OR 3.19, 95% CI 1.38–7.34%), while the estimates for postsurgical onset (1.61, 0.58–4.51%) and relief after a block (2.43, 0.58–10.12%) were attenuated and no longer significant. This bears directly on the robustness of the two associations most vulnerable to documentation bias, and indicates that the block-response association in particular is not stable across analytic choices.

## DISCUSSION

In this large single-center cohort of adults with pudendal neuralgia, three findings stand out.

First, a confirmed postsurgical onset was less common than text screening implied. Screening flagged nearly half the cohort as dating their pain to an operation; chart verification confirmed a named operation preceding the pain in 13.8%, fewer than the proportion with documented entrapment features. Iatrogenic onset is therefore not the leading identifiable mechanism in this cohort, and studies that rely on automated extraction alone are liable to conclude otherwise.

Second, neuromodulation was documented far more often than it was delivered. A device was implanted in 8.0% of the cohort, against the 28.6% in whom neuromodulation appeared in the record, and most of the difference consists of patients in whom the treatment was discussed, offered, or planned but never carried out. Of the devices that were implanted, five in

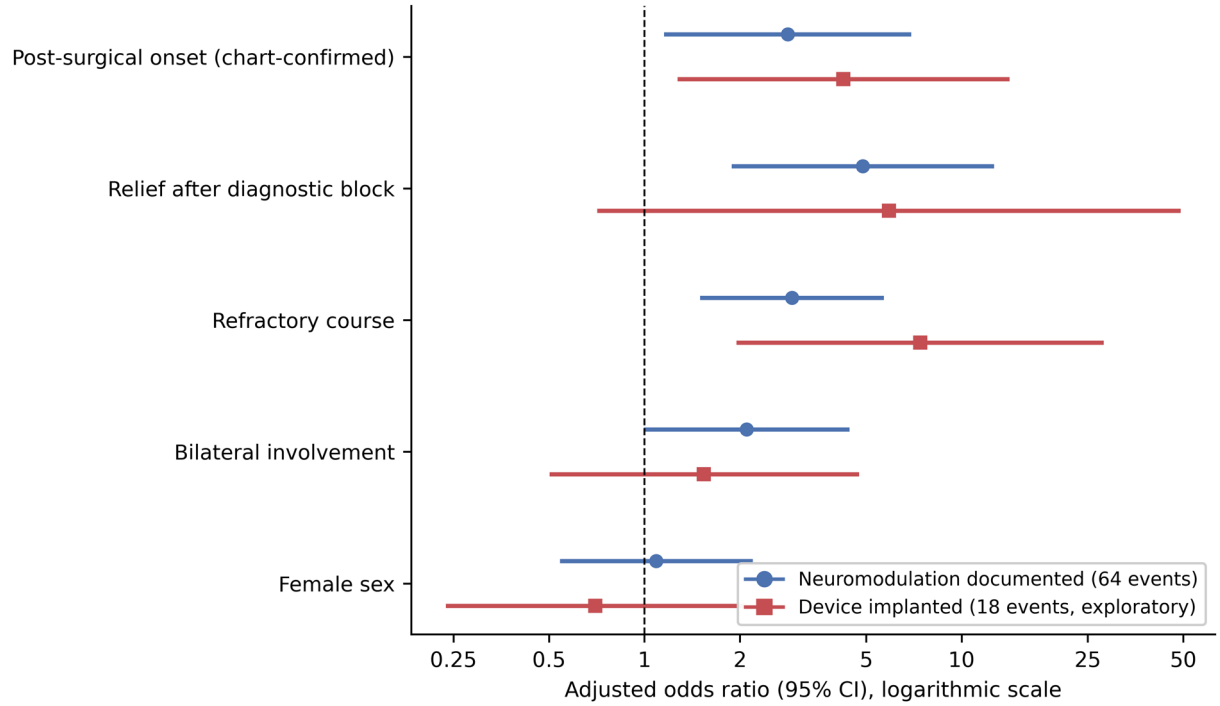



**Fig. 3** Factors associated with escalation to neuromodulation under two outcome definitions. Adjusted odds ratios (points) with 95% confidence intervals (bars) from two multivariable logistic-regression models, plotted on a logarithmic scale; the vertical dashed line marks an odds ratio of 1. Blue circles denote the primary outcome, neuromodulation documented in the record (64 events); red squares denote the exploratory secondary outcome, permanent device implanted (18 events). A refractory course and a chart-confirmed postsurgical onset were associated with both outcomes. Documented relief after a diagnostic block was associated with the primary outcome but its confidence interval crossed unity for implantation

six were spinal cord stimulators and none was a dedicated pudendal nerve stimulator.

Third, the factors associated with escalation depend on which end of that pathway is examined. A refractory course was associated with both documented neuromodulation and actual implantation, and was the strongest factor for implantation. A confirmed postsurgical onset was likewise associated with both. Documented relief after a diagnostic block was strongly associated with neuromodulation being documented, but its association with implantation did not reach significance and it did not survive restriction to the best-documented records. Sex was not associated with either outcome.

The clinical picture we observed is consistent with the classical description of pudendal neuralgia: neuropathic perineal pain that is characteristically worse when sitting, with a female predominance [3–5]. Pelvic surgery is an established cause of pudendal nerve injury or entrapment, including after mesh and sling procedures [4, 5, 12, 16], and our confirmed cases are consistent with that literature in their distribution: hysterectomy, colorectal surgery, endometriosis laparoscopy, prolapse, and hernia repair accounted for most of them. What our data do not support is the stronger claim that iatrogenic pudendal neuralgia is the leading identifiable mechanism in a specialty pain service. At 13.8%, it is less frequently confirmed than documented entrapment features. It is, however, disproportionately associated with escalation: confirmed postsurgical patients reached a permanent device more often than other patients (unadjusted OR 3.62, 95% CI 1.25–10.51%). The appropriate inference is therefore about trajectory rather than frequency, and the practical

implications for consent around pelvic surgery and for early referral of postoperative perineal neuropathic pain rest on that narrower footing.

A separate question is which findings distinguished pudendal nerve entrapment from pudendal neuralgia without entrapment. In this service, that distinction was made descriptively rather than by protocol. Entrapment was recorded in 54 patients (24.1%) when the treating clinician documented entrapment or Alcock-canal involvement, or when the patient was referred for or underwent decompressive surgery on that indication. The features that accompanied such documentation were pain reproduced or worsened by prolonged sitting and relieved by standing or lying, tenderness on palpation at the ischial spine or within Alcock's canal, and relief after a block placed at that level. Neurophysiological testing, including pudendal nerve terminal motor latency, was not routinely performed and was recorded in only a small minority, and magnetic resonance neurography was not available for most of the study period. The distinction in these records therefore rests on clinical pattern and block response rather than on imaging or electrophysiological confirmation, and we would not claim that the 54 patients represent anatomically verified entrapment. This is a limitation of routine-care data rather than of the classification itself, and it is one reason the Nantes criteria, which were designed for exactly this difficulty, should be applied prospectively.

The finding that a documented block response was associated with neuromodulation mirrors the logic of the Nantes framework, in which a response to an anesthetic pudendal block both supports the diagnosis and identifies the pain generator [3]. Diagnostic and therapeutic pudendal blocks, whether ultrasound-, fluoroscopy-, or computed-tomography-guided, are well established [14–17], and infiltration response has long been used, alongside clinical and neurophysiological assessment, to select patients for surgical decompression [6, 15]. Our results indicate that, in contemporary practice, a positive block increasingly functions as a gateway to neuromodulation rather than to surgery. While randomized data show that adding a corticosteroid to the local anesthetic confers no benefit [18], and prospective uncontrolled data suggest that pulsed radiofrequency can provide durable relief [19]; neuromodulation, whether spinal cord, dorsal root ganglion, sacral, or dedicated pudendal nerve stimulation, has emerged as a favored option for refractory perineal pain [20, 22–26]. In our cohort spinal cord, stimulation was the modality used in 14 of 18 implanted patients, sacral neuromodulation in 2, dorsal root ganglion stimulation in 1 and a sciatic peripheral nerve stimulator in 1, with no dedicated pudendal nerve stimulator implanted at any point in 16 years.

How a specific modality was selected can be described only in part, because the records support an account of what was chosen but not a reconstruction of why. No written selection algorithm existed in this service during the study period, and the notes rarely set out the reasoning for choosing one target over another. What can be stated is the pattern. Spinal cord stimulation predominated throughout, and in the four patients whose operative detail was explicit, the leads were positioned to cover the sacral or lower lumbar segments, in one case introduced through the sacral hiatus. Dorsal root ganglion stimulation appeared only late in the study period and in a patient who had first undergone a spinal cord stimulation trial, which is consistent with its adoption as a second-line target after a failed or inadequate conventional trial rather than as a primary choice. Sacral neuromodulation was used in two patients, both with urinary symptoms recorded alongside the pain. The absence of dedicated pudendal nerve stimulation, despite its description in the literature [23–26], most plausibly reflects local availability and operator familiarity rather than a considered decision against the target, but we cannot demonstrate this from the record. Readers should therefore treat the modality distribution as a description of one service's practice, shaped by availability and experience, and not as evidence of comparative effectiveness between targets. A prospective registry recording the indication for each target would be needed to establish how modality should be selected. The relatively modest use of surgical decompression, despite its established long-term efficacy in entrapment [6, 7, 13], is consistent with this shift toward reversible neuromodulatory

approaches, as is the observation that a proportion of decompressed patients have persistent pain and require revision surgery [21].

This link should be read in a clinical context, because the relationship is partly circular. A documented block response and a refractory course are, in everyday practice, part of the reasoning that leads to offering neuromodulation, so their association with it partly reflects how patients are selected rather than an independent biological effect. Our data now allow that circularity to be partly separated. The block-response association was strong against documented neuromodulation, weaker and nonsignificant against actual implantation, and did not survive restriction to the best-documented records. That pattern is what one would expect if a recorded block response travels with the decision to consider a device rather than with the decision to implant one. A documented block response should not therefore be read as the factor that identifies patients who reach neuromodulation, and we view these findings as describing the interventional pathway rather than predicting escalation.

The pharmacological pattern, with gabapentinoids, tricyclics, and duloxetine as mainstays, accords with guideline first-line therapy for neuropathic pain [27]. The substantial use of opioids, in about one third of patients, is notable given their limited role in chronic noncancer pain and current prescribing guidance [28], and the frequent use of medical cannabis reflects its increasing though still weakly evidenced adoption for chronic pain [29]. Together these observations reinforce a mechanism-based, stepped model in which nonopioid pharmacotherapy, pelvic-floor rehabilitation, and image-guided diagnostic blocks are used early, with a positive block directing selected patients, particularly those with postsurgical and refractory disease, toward neuromodulation.

Although women outnumbered men in this cohort, as expected from the literature, female sex was not independently associated with escalation to neuromodulation once etiology, block response, and refractoriness were taken into account. This suggests that the female predominance of pudendal neuralgia reflects who develops the condition rather than who receives its most invasive treatments, and it argues against sex-based assumptions when considering referral for neuromodulation.

## Limitations

This study has the limitations inherent to a retrospective, single-center analysis of routine medical records, and two of them can be quantified here.

The cohort is clinician-defined. The diagnosis rested on the treating clinicians' assessment rather than on formal application of the Nantes criteria, which could not be reconstructed retrospectively. The frequencies we report reflect what was recorded in practice and under-represent features that are less consistently written down; the figures for sitting-related and burning pain are best read as documentation rates rather than prevalences.

Text screening over-identified both of the variables that were subsequently verified. For the postsurgical variable, screening indicated 46.4% and chart verification confirmed 13.8%; for neuromodulation, screening indicated 28.6% and chart verification confirmed an implanted device in 8.0%. The size of these discrepancies bears on how studies of this condition that rely on automated extraction should be read. One asymmetry remains and cannot be removed: chart readjudication was applied only to records that screening had flagged, so it can remove false positives but cannot detect patients whose postsurgical onset or implanted device was never captured by the screening terms. The confirmed figures are therefore lower bounds, and the true values lie somewhere between the confirmed and the screened estimates, probably much closer to the former.

Treatment response was based on documented clinical improvement rather than a standardized, quantitative outcome measure; no patient-reported outcome instrument was available, and no numerical threshold for a block response could be applied. Absence of documentation was coded as absence of the feature throughout.

The exploratory model of permanent implantation contains 18 events against five covariates and is underpowered; its estimates are unstable

and are presented for comparison with the primary model rather than as independent findings. Temporal sequence between an operation and the onset of pain does not establish iatrogenic nerve injury, and the confirmed postsurgical variable should be read accordingly.

As in any observational study, the associations we describe reflect patterns of care rather than proven cause and effect, and a documented block response and a refractory course are, in part, the clinical reasons neuromodulation is offered, so their association with it is subject to confounding by indication. Patients presenting later in the study period had less opportunity to escalate, which we addressed by adjustment for treatment era but cannot fully exclude. Finally, as a single tertiary referral center, our cohort is likely enriched for complex and refractory disease, and the proportions reported here should not be read as population estimates.

These considerations notwithstanding, the study draws on a large sample for an uncommon condition, spans a long study period, and captures presentation, etiology, and the full range of treatments in detail, offering a real-world picture of how pudendal neuralgia is diagnosed and managed in a specialty pain service.

## CONCLUSIONS

In routine pain-clinic practice, pudendal neuralgia is female-predominant and managed along an interventional pathway, but a permanent neuromodulation device is implanted in far fewer patients than the frequency with which neuromodulation is discussed, and spinal cord stimulation rather than dedicated pudendal stimulation is what is actually used. A postsurgical onset, when confirmed against the chart, is less common than text screening suggests and less common than documented entrapment features, yet is disproportionately associated with reaching a device. A refractory course was the factor most consistently associated with escalation at every level of the pathway. Because the analysis is observational, these findings describe how such patients are managed rather than proving cause. A prospective study, ideally multicenter and with standardized application of the Nantes criteria, a prespecified definition of block response, and patient-reported outcomes, is needed to establish which factors genuinely predict escalation and which modality suits which patient.

## ACKNOWLEDGEMENTS

We thank the participants of the study. The authors thank the patients whose de-identified records made this study possible.

***Author Contributions.*** Conceptualization: Paz Kelmer, Shachar Zion Shemesh, Lior Ungar. Methodology: Paz Kelmer, Shachar Zion Shemesh, Lior Ungar. Formal analysis and investigation: Paz Kelmer, Shachar Zion Shemesh, Philip Rossovsky, Yotam Hadari. Medical-record review and chart adjudication: Paz Kelmer, Shachar Zion Shemesh, Philip Rossovsky. Data curation: Paz Kelmer, Shachar Zion Shemesh, Philip Rossovsky, Yotam Hadari. Writing—original draft preparation: Paz Kelmer, Shachar Zion Shemesh. Writing—review and editing: Philip Rossovsky, Yotam Hadari, Itay Goor-Aryeh, Lior Ungar. Clinical interpretation: Itay Goor-Aryeh, Lior Ungar. Supervision: Lior Ungar. All authors read and approved the final manuscript. All named authors meet the International Committee of Medical Journal Editors (ICMJE) criteria for authorship of this article, take responsibility for the integrity of the work as a whole, and have given their approval for this version to be published.

***Funding.*** No funding or sponsorship was received for the conduct of this study. The Rapid Service Fee for publication of this article was funded by the Sheba Fund for Health Services and Research, Sheba Medical Center, Ramat Gan, Israel.

***Data Availability.*** The datasets generated during and/or analyzed during the current study are not publicly available because they contain information that could compromise the privacy of individual patients, but de-identified data are

available from the corresponding author on reasonable request and subject to institutional approval.

### *Declarations*

***Conflict of Interest.*** Paz Kelmer, Shachar Zion Shemesh, Philip Rossovsky, Yotam Hadari, Itay Goor-Aryeh, and Lior Ungar declare that they have no competing interests.

***Ethical Approval.*** The study was approved by the institutional Helsinki (research ethics) committee of the Sheba Medical Center, Tel HaShomer, Israel (approval no. SMC-8148-21; Israeli Ministry of Health no. 202121003), with a waiver of informed consent for this retrospective analysis of de-identified records. It was performed in accordance with the Declaration of Helsinki of 1964 and its later amendments. No individually identifiable data are presented, and consent for publication was therefore not applicable. Patients were not involved in the design, conduct, or reporting of this retrospective record-based study.

***Open Access.*** This article is licensed under a Creative Commons Attribution-NonCommercial 4.0 International License, which permits any non-commercial use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc/4.0/.

## REFERENCES

1. Ahangari A. Prevalence of chronic pelvic pain among women: an updated review. Pain Physician. 2014;17(2):E141–7.
2. Schaeffer AJ. Epidemiology and evaluation of chronic pelvic pain syndrome in men. Int J Antimicrob Agents. 2008;31(Suppl 1):S108–11.
3. Labat JJ, Riant T, Robert R, Amarenco G, Lefaucheur JP, Rigaud J. Diagnostic criteria for pudendal neuralgia by pudendal nerve entrapment (Nantes criteria). Neurourol Urodyn. 2008;27(4):306–10.
4. Stav K, Dwyer PL, Roberts L. Pudendal neuralgia. Fact or fiction? Obstet Gynecol Surv. 2009;64(3):190–9.
5. Hibner M, Desai N, Robertson LJ, Nour M. Pudendal neuralgia. J Minim Invas Gynecol. 2010;17(2):148–53.
6. Robert R, Prat-Pradal D, Labat JJ, et al. Anatomic basis of chronic perineal pain: role of the pudendal nerve. Surg Radiol Anat. 1998;20(2):93–8.
7. Popeney C, Ansell V, Renney K. Pudendal entrapment as an etiology of chronic perineal pain: diagnosis and treatment. Neurourol Urodyn. 2007;26(6):820–7.
8. Filler AG. Diagnosis and treatment of pudendal nerve entrapment syndrome subtypes: imaging, injections, and minimal access surgery. Neurosurg Focus. 2009;26(2):E9.
9. Scholz J, Finnerup NB, Attal N, et al. The IASP classification of chronic pain for ICD-11: chronic neuropathic pain. Pain. 2019;160(1):53–9.
10. Treede RD, Rief W, Barke A, et al. Chronic pain as a symptom or a disease: the IASP classification of chronic pain for ICD-11. Pain. 2019;160(1):19–27.
11. Cardaillac C, Delga B, Thubert T, et al. Description and classification of postpartum chronic pain: a multicentric prospective study. J Gynecol Obstet Hum Reprod. 2020;49(9):101769.
12. Sancak EB, Avci E, Erdogru T. Pudendal neuralgia after pelvic surgery using mesh: case reports and laparoscopic pudendal nerve decompression. Int J Urol. 2016;23(9):797–800.
13. Beco J, Climov D, Bex M. Pudendal nerve decompression in perineology: a case series. BMC Surg. 2004;4:15.
14. Fanucci E, Manenti G, Ursone A, et al. Role of interventional radiology in pudendal neuralgia: a

description of techniques and review of the literature. Radiol Med. 2009;114(3):425–36.

15. Mamlouk MD, van Sonnenberg E, Dehkharghani S. CT-guided nerve block for pudendal neuralgia: diagnostic and therapeutic implications. Am J Roentgenol. 2014;203(1):196–200.

16. Wadhwa V, Scott KM, Rozen S, Starr AJ, Chhabra A. CT-guided perineural injections for chronic pelvic pain. Radiographics. 2016;36(5):1408–25.

17. Ly J, Scott K, Xi Y, Ashikyan O, Chhabra A. Role of 3 Tesla MR neurography and CT-guided injections for pudendal neuralgia: analysis of pain response. Pain Physician. 2019;22(4):E333–44.

18. Labat JJ, Riant T, Lassaux A, et al. Adding corticosteroids to the pudendal nerve block for pudendal neuralgia: a randomised, double-blind, controlled trial. BJOG. 2017;124(2):251–60.

19. Masala S, Calabria E, Cuzzolino A, Raguso M, Morini M, Simonetti G. CT-guided percutaneous pulse-dose radiofrequency for pudendal neuralgia. Cardiovasc Interv Radiol. 2014;37(2):476–81.

20. Levesque A, Bautrant E, Quistrebert V, et al. Recommendations on the management of pudendal nerve entrapment syndrome: a formalised expert consensus. Eur J Pain. 2022;26(1):7–17.

21. Hibner M, Castellanos ME, Drachman D, Balducci J. Repeat operation for treatment of persistent pudendal nerve entrapment after pudendal neurolysis. J Minim Invas Gynecol. 2012;19(3):325–30.

22. Gregory NS, Terkawi AS, Prabhakar NK, Tran JV, Salmasi V, Hah JM. Peripheral nerve stimulation for pudendal neuralgia: a technical note. Pain Med. 2020;21(Suppl 1):S51–5.

23. Hunter CW, Falowski S. Neuromodulation in treating pelvic pain. Curr Pain Headache Rep. 2021;25(2):9.

24. Valovska A, Peccora CD, Philip CN, Kaye AD, Urman RD. Sacral neuromodulation as a treatment for pudendal neuralgia. Pain Physician. 2014;17(5):E645–50.

25. Buffenoir K, Rioult B, Hamel O, Labat JJ, Riant T, Robert R. Spinal cord stimulation of the conus medullaris for refractory pudendal neuralgia: a prospective study of 27 consecutive cases. Neurourol Urodyn. 2015;34(2):177–82.

26. Ghorayeb JH, Chitneni A, Rupp A, Parkash A, Abd-Elsayed A. Dorsal root ganglion stimulation for the treatment of chronic pelvic pain: a systematic review. Pain Pract. 2023;23(7):838–46.

27. Finnerup NB, Attal N, Haroutounian S. Pharmacotherapy for neuropathic pain in adults: a systematic review and meta-analysis. Lancet Neurol. 2015;14(2):162–73.

28. Dowell D, Ragan KR, Jones CM, Baldwin GT, Chou R. CDC clinical practice guideline for prescribing opioids for pain – United States, 2022. MMWR Recomm Rep. 2022;71(3):1–95.

29. Busse JW, Vankrunkelsven P, Zeng L, et al. Medical cannabis or cannabinoids for chronic pain: a clinical practice guideline. BMJ. 2021;374:n2040.